

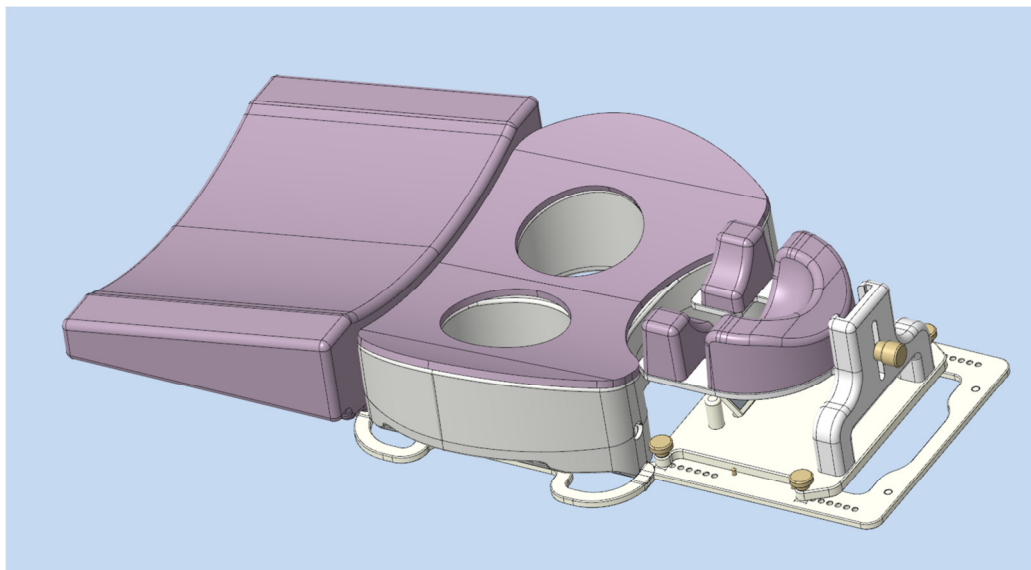
Service Manual

for

16Ch Diagnostic Breast Coil - P-H16LE-030-01893

to be operated on

GE 3.0 T MR Systems



3.0T GEHC Part no.: 5772248



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1 Important Information

ПРЕДУПРЕЖДЕНИЕ (BG)	Това упътване за работа е налично само на английски език. - Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод. - Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа. - Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.
警告 (ZH-CN)	本维修手册仅提供英文版本。 - 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 - 未详细阅读和完全理解本维修手册之前，不得进行维修。 - 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。
警告 (ZH-HK)	本服務手冊僅提供英文版本。 - 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。 - 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。 - 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。
警告 (ZH-TW)	本維修手冊僅有英文版。 - 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。 - 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。 - 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。
UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. - Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. - Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. - Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for tekniker, operatør eller patient.

WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.
WARNING (EN)	This service manual is available in English only. - If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services. - Do not attempt to service the equipment unless this service manual has been consulted and is understood. - Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles - Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. - Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. - Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. - Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. - Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. - Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. - Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. - Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. - Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. - Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. - Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. - Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά. - Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης. - Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις. - Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

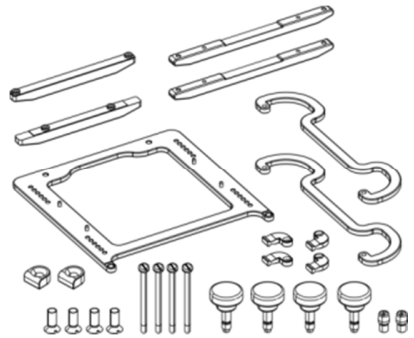
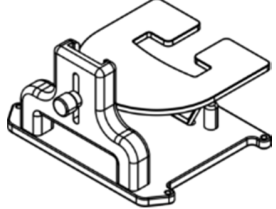
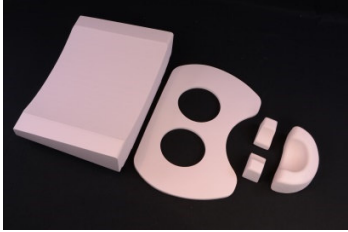
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. - Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése. - Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. - Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
AÐVÖRUN (IS)	Þessi þjónustuhandbók er aðeins áánleg á ensku. - Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu. - Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. - Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. - Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. - Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. - Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.
警告 (JA)	このサービスマニュアルには英語版しかありません。 - サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 - このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 - この警告に従わない場合、サービスを担当される方、操作員あるいは患者・・・さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객님의 책임입니다. 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai anglū valodā. - Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. - Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. - Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.
ĮSPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. - Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. - Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. - Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. • Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.
AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. • Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar este manual noutra idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. • Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. • Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. • Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.

UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. . • Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
OPOZORILO (SL)	Ta servisni priročnik je na voljo samo v angleškem jeziku. • Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod. • Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli. • Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. • Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. • Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. • Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

2 Introduction

2.1 Shipping Contents

Description	GEHC Part #	Rapid Part No.	Illustration
3.0T 16Ch Diagnostic Breast Coil with cable	5772248-2	P-H16LE-015-01630	
16Ch Diagnostic Breast Coil positioning parts set	5772248-3	ZUB-01858	
16Ch Diagnostic Breast Coil head rest framework	5772248-4	ZUB-01859	
16Ch Diagnostic Breast Coil cushion set	5772248-5	ZUB-01860	
CD containing user manual (all languages)	N/A	RT004808	N/A
3.0 T Foreign Language Sticker Sheet	N/A	RT004812	N/A
eIFU Leaflet	N/A	RT004810	N/A

16ch Diagnostic Breast Coil service manual	N/A	RT004811	N/A
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2.2 Environment Storage Requirement

Store the coil and accessories in an air conditioned scan room or equipment room until the GE field engineer can perform installation. The environmental condition limits are included below.

	<i>Operating Conditions</i>	<i>Transport & Storage Conditions</i>
<i>Temperature Range</i>	+15°C to +24°C +59°F to +75.2°F	-25°C to +60°C -13°F to +140°F
<i>Relative Humidity</i>	30 % to 80 % RH	5 % to 95 % RH
<i>Air Pressure</i>	70 kPa - 107 kPa	n/a

3 Coil Installation

3.1 Plug label and Mirror protection

3.1.1.1 List of required parts

Description	Quantity	Illustration
Foreign Language Sticker Sheet	1	

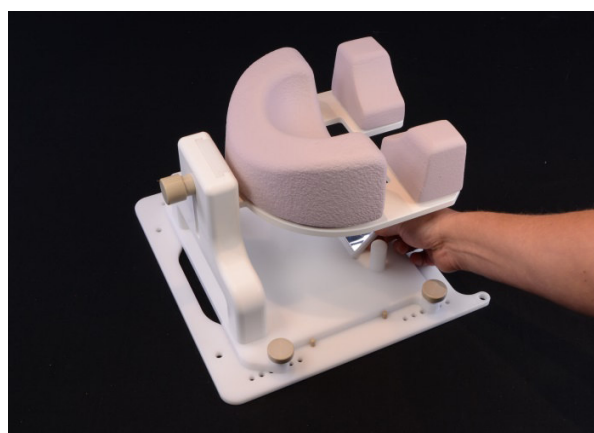
3.1.1.2 Installation instruction



Step 1: Select the sticker according to the national language



Step 2: Paste the sticker to the empty field of the connector



Step 3: Peel of the protective film from the mirror of the head rest

3.2 Adapting the coil to the patient table

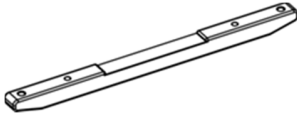
The housing of the 16Ch Diagnostic Breast Coil and the head rest framework are adapted to the patient table of the individual MR system by positioning parts. The following subsections give specific instructions for the necessary adjustments in correspondence to the MR system on site.

Tools are not included the package. Use appropriate screw drivers to avoid wearing out the screw heads. Do not use excessive force when tightening the screws.

3.2.1 Curved Table

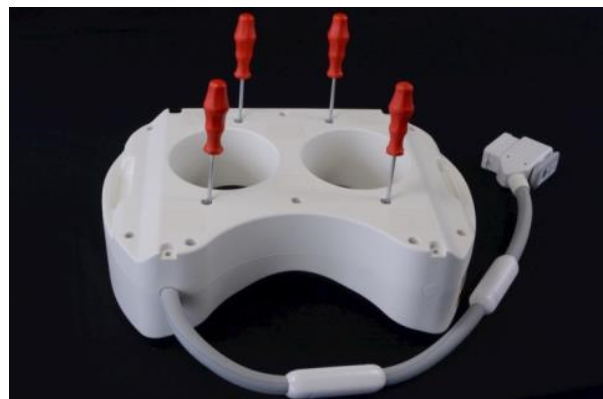
This is the assembly instruction for the Curved Table. Not all of the enclosed parts are required. Select the parts shown in the list below only.

3.2.1.1 List of required parts

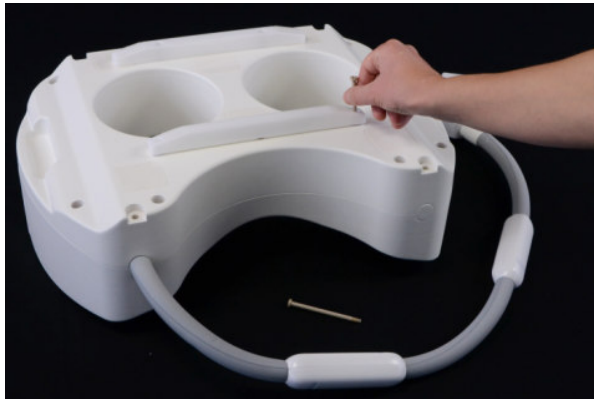
Description	Quantity	Illustration
Elbow bracket	4	
Short stand with inner mounting holes	1	
Short stand with outer mounting holes	1	
Long stand for head rest	2	
Long slotted flat headcheese head screw	4	
Short slotted countersunk screw	4	
knurled Thumb screw	4	

3.2.1.2 Installation instruction

The following steps show how to adapt the coil and the head rest to the Curved Table.



Step 1: Turn the coil upside down



Step 2: Unscrew the four screws marked by the screwdrivers in the image above.



Step 3: Align the two short stands in the holes where you have removed the screws. Insert the four long slotted flat head screws through the holes of the stands.

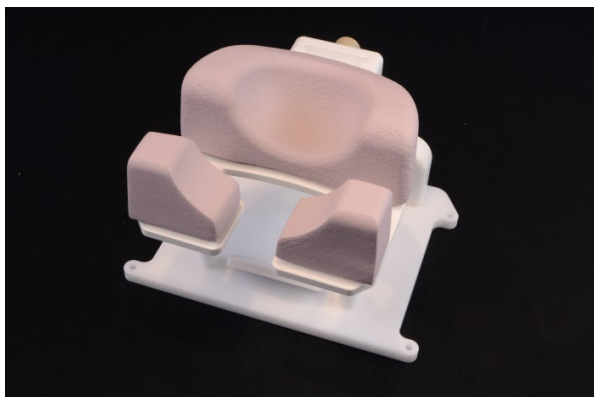


Step 4: Tighten the four screws.



Step 5: In each corner, place a short slotted countersunk screw and an elbow bracket. The bracket points towards the side of the coil. Tighten the screws.

Step 6: Continue until all four brackets are mounted as shown above.



Step 7: Take the head rest and turn it upside down.

Step 8: Take the two long stands and the four knurled thumb screws



Step 9: Tighten all four screws by hand to finish the assembly

3.2.2 GEM Table (narrow flat table)

This is the assembly instruction for the GEM Table. Not all of the enclosed parts are required. Please select the parts shown in the list below only.

3.2.2.1 List of required parts

Description	Quantity	Illustration
Elbow bracket	4	
Short slotted countersunk screw	4	

3.2.2.2 Installation instruction



Step 1: Turn the coil upside down



Step 2: In each corner, place a short slotted countersunk screw and an elbow bracket. The bracket points towards the side of the coil. Tighten the screws.

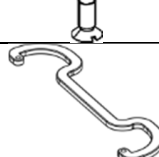


Step 3: Continue until all four brackets are mounted as shown above.

3.2.3 Fixed Table (wide flat table)

This is the assembly instruction for the wide Fixed Table. Not all of the enclosed parts are required. Please select the parts shown in the list below only.

3.2.3.1 List of required parts

Description	Quantity	Illustration
Short slotted countersunk screw	4	
Wide table bracket	2	

3.2.3.2 Installation instruction



Step 1: Turn the coil upside down



Step 2: Take the four short slotted countersunk screws and the two wide table brackets. Place them to the mounting slots and tighten the screws.

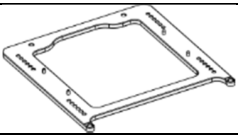


Step 3: Continue until both brackets are mounted.

3.2.4 PET/MR Table

This is the assembly instruction for the PET/MR Table. Not all of the enclosed parts are required. Please select the parts shown in the list below only.

3.2.4.1 List of required parts

Description	Quantity	Illustration
Alignment cone	2	
PET head positioning frame	2	
Mounting slot cover	2	
Short slotted countersunk screw	2	
Thumb screw	4	

3.2.4.2 Installation instruction



Step 1: Turn the coil upside down



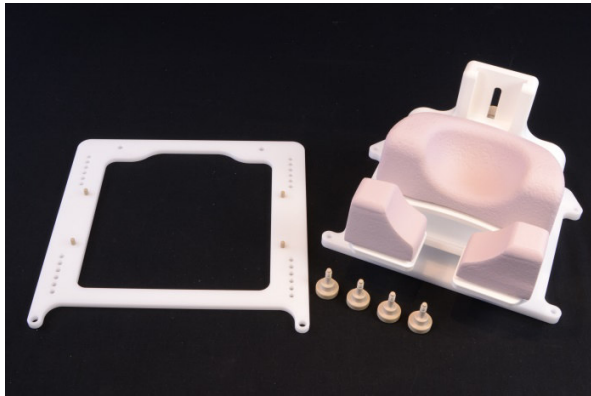
Step 2: Screw one of the two “alignment cones” into the thread near the cable outlet..



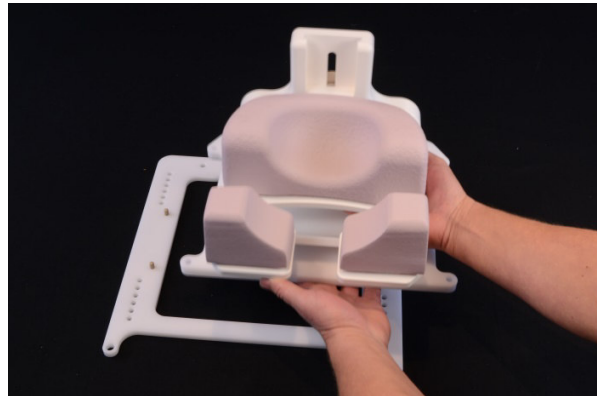
Step 3: Continue with the second “alignment cone” on the opposite side as shown above.



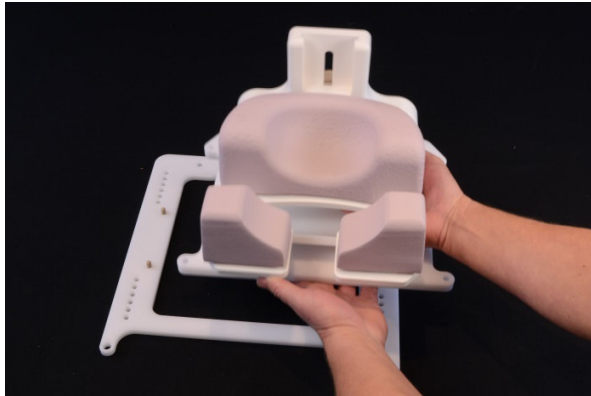
Step 4: Cover the remaining two slots using “mounting slot covers” and a short slotted countersunk screw.



Step 5: Take the PET positioning frame and the thumb screw



Step 6: Put the head rest onto the frame



Step 7: Tighten the four thumb screws

3.3 Performance / Quality Assurance (QA)

We recommend regular verification of the device's proper function by performing the Coil Quality Assurance test utilizing GE Multi-Coil Quality Assurance (MCQA) Tool.

3.3.1 MCQA setup

3.3.1.1 List of required parts

Description	Quantity	Illustration
16Ch Diagnostic Breast Coil	1	
Phantom holder foam ring	1	
Phantom GEHC part no. 5342679-2 (Large Cylindrical Unified Phantom, SiOil)	1	

3.3.1.2 Phantom positioning

The patient position is prone, feet forward. This results in the designation of “left” and “right” as shown in Figure 1.



Figure 1: MCQA setup –

left: Phantom placed inside the left cavity of the coil;

right: Phantom placed inside the right cavity of the coil.

Each cavity of the coil will be tested separately. Therefore the MCQA test must be run twice to test a single coil.

First place the foam holder in one of the cavities of the coil. The foam holder is elliptic; check its orientation for an easy fit. Place the phantoms inside the holder. The phantom shall touch the surface of the patient table.

3.3.2 Description of the MCQA test procedure

3.3.2.1 MR system's calibration

All RF coil related tests have to be performed on a system that is calibrated. The command "install_in_spec" and especially EPIWP (White Pixel from install in spec) is to result in PASS.

3.3.2.2 Initiating MCQA

- 1) From Common Service Desktop (CSD), go to **Service Browser** and select **[Image Quality]** "**Multi-Coil QA Tool**" and then "**Click here to start this tool**" as shown Figure 2.

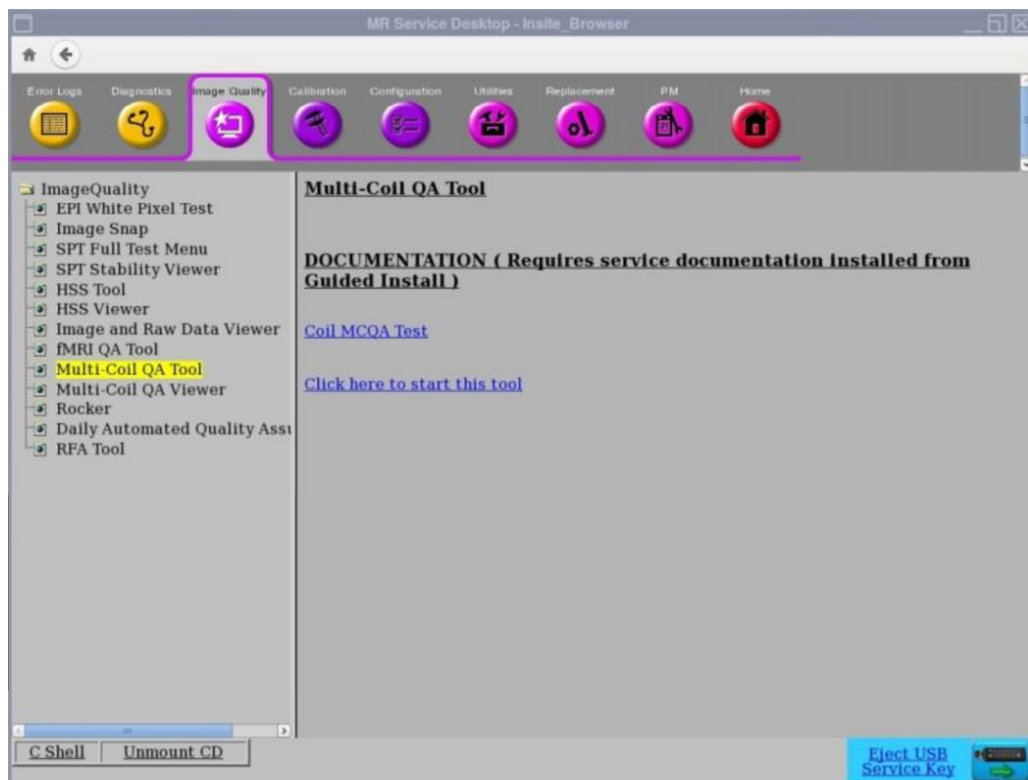


Figure 2: Multi-Coil QA Tool – "Click here to start this tool"

Note: If a "No valid MCR-V (or MCR2/3)" warning (Figure 3) pops up click **[Yes]** and proceed with test. MCR-V diagnostics must be run before turning over system to customer.



Figure 3: Warning "No valid MCR-V (or MCR2/3)"

The field “current coil” will be automatically filled in (Figure 4) with the coil name, based on the CoilID of the coil connected to the LPCA. Choose entry ending as “Left_Array” or “Right_Array” for the test with the phantom placed inside the left or the right cavity of the coil. Enter the serial number of the coil being tested in the field “Coil Serial #”.

- 2) Click **[Start]** to begin the automated test as shown in Figure 4. The test may take from 3-5 minutes depending on the MR system.

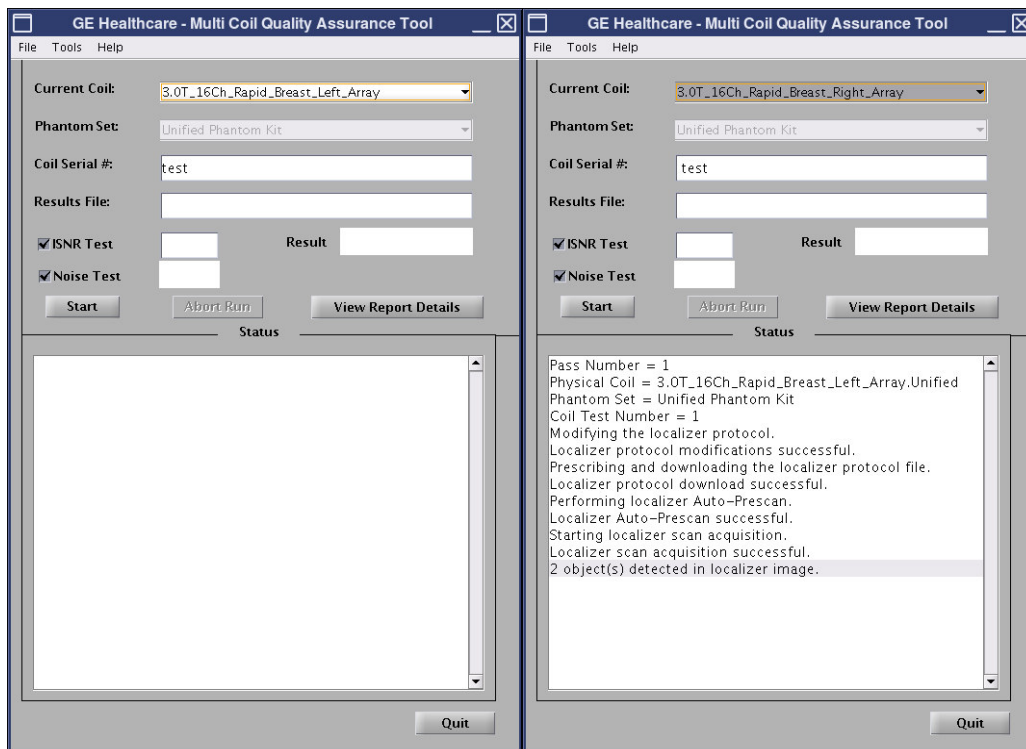


Figure 4: Selection of Current Coil for tests on the left or the right cavity of the coil.

- 3) Upon start-up, a message stating “Phantom placement and coil landmarking are critical for repeatable results” will appear. If the landmark has been set correctly and there are no air bubbles in the phantom, click **[Yes]** to continue. (Figure 5)

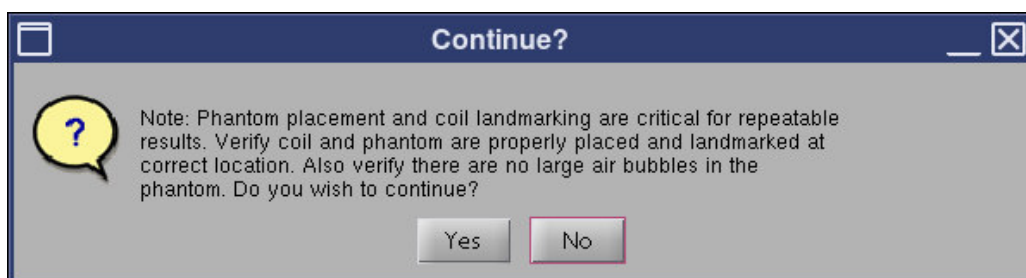


Figure 5: Warning Phantom placement & landmarking

Note: The Status window of the MCQA Tool GUI will continuously update to give information on what the tool is doing at any point in time. A progress bar (Figure 6) will appear, showing approximate total test time, elapsed time and percent complete.

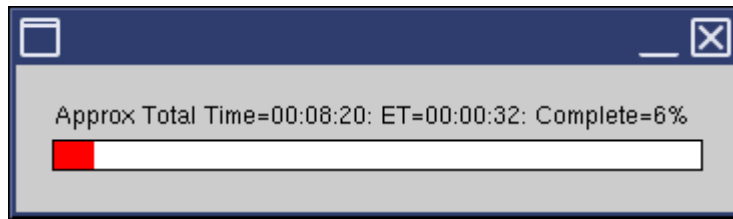


Figure 6: Status window

Note: The test protocol might display further (error) messages. In this case the test is considered **FAIL**. See below for corrective actions.

After the test has been completed, test results are displayed on the screen (Figure 7).

The PASS/FAIL status shows **PASS** if all coil elements are functioning properly.

The MCQA Tool GUI displays **FAIL**, e.g. for one of the following reasons:

- Incorrect positioning/placement of the phantom (cf. Figure 8)
- Incorrect phantom used for the test (cf. Figure 9)
- Bad Coil Element (contact service)

Refer to section 5.2 Failed MCQA if the MCQA test fails.

More information on the MCQA test can be found on MR service methods DVD or website via the path: Troubleshooting -> System -> Multi-Coil Quality Assurance Tool

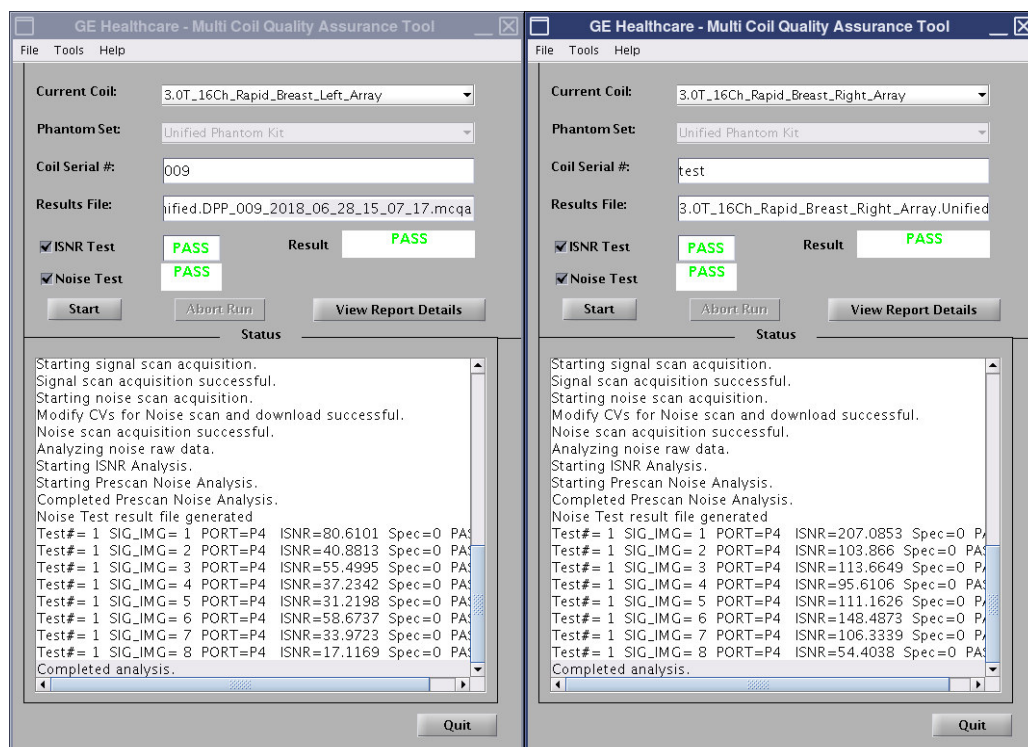


Figure 7: Tests results displayed on the screen –

In this example all tests were passed by the left and the right array half

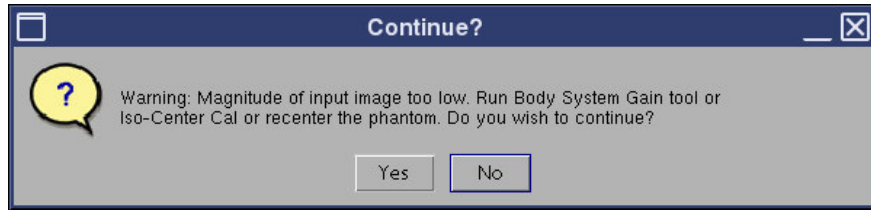


Figure 8: Error message related to wrong selection of the cavity. Make sure the “Current coil” is selected corresponding the the placement of the phantom.

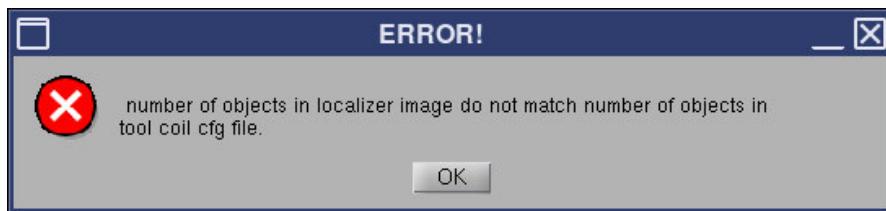


Figure 9: Error message related to wrong phantom. Place only one phantom in one of the cavities. Tests with two phantoms simultaneously are not possible.

- 4) Perform the test for the other cavity of the coil. The test for both cavities must result in **PASS**.
- 5) Click on **[Quit]** button to exit MCQA Tool.

3.3.2.3 Using MCQA Viewer

In case if the results are to be viewed at a later stage follow the below steps:

- 1) In the MCQA Tool window select File, Open Results File and select the desired coil results file select **[View Report Details]** to review the results.

Note: The Results Viewer will open as shown in Figure 10. The Results file name and Pass/Fail Results shown on the tool GUI will also be listed across the top of the viewer.

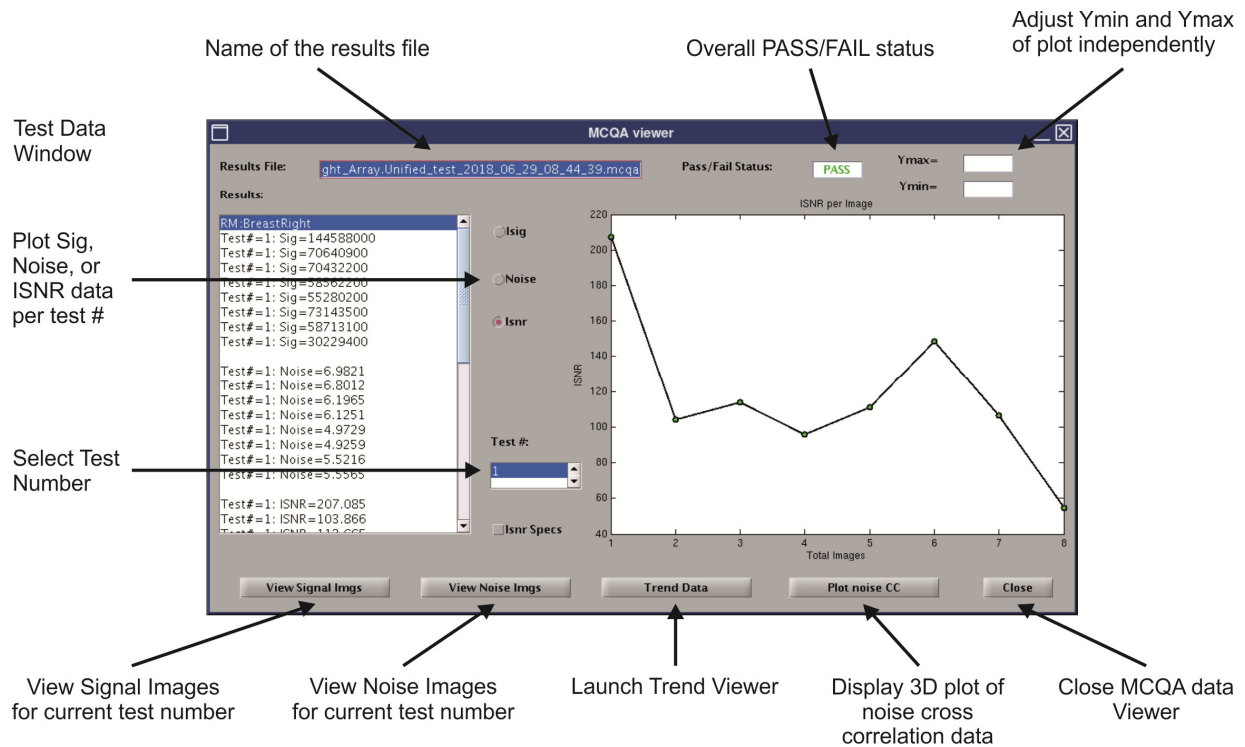


Figure 10: MCQA Viewer

- 2) Select the ISNR option and the ISNR Specs check box in the middle portion of the Results Viewer to view the results.

3.3.3 Execution of the MCQA test procedure

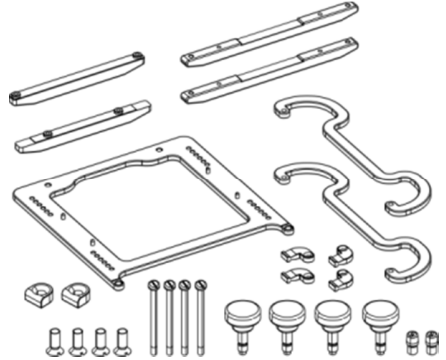
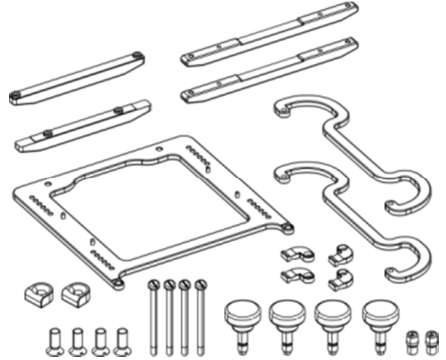
Run the MCQA test as described above for the left and the right cavity. Position the phantom accordingly. Each MCQA test has to result in **PASS**.

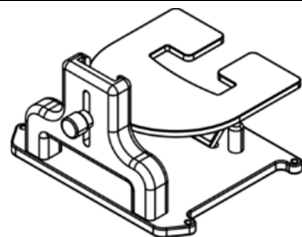
Personal Requirements	Preliminary Reqs	Procedure	Finalization
1 person	0 min	20 mins	0 min

Test ID	Parameter Description	Expected Result
"left"	"install_in_spec" passed	PASS

4 Field Replaceable Units (FRUs)

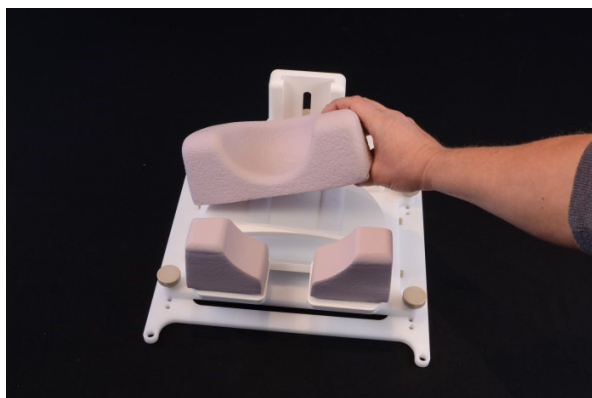
4.1 List of Field Replaceable Units

Description	GEHC Part #	Rapid Part No.	Illustration
1.5T 16Ch Diagnostic Breast Coil with cable	5772249-2	P-H16LE-015-01811	
3.0T 16Ch Diagnostic Breast Coil with cable	5772248-2	P-H16LE-030-01630	
1.5T 16Ch Diagnostic Breast Coil positioning parts set	5772249-3	ZUB-01857	
3,0T 16Ch Diagnostic Breast Coil positioning parts set	5772248-3	ZUB-01858	

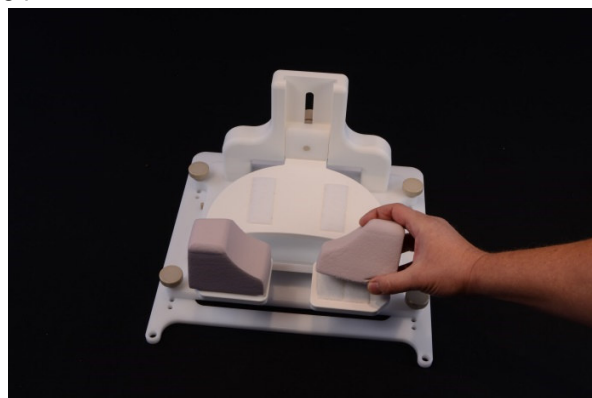
16Ch Diagnostic Breast Coil head rest framework	5772248-4	ZUB-01859	
16Ch Diagnostic Breast Coil cushion set	5772248-5	ZUB-01860	

4.2 Replacement of the cushions

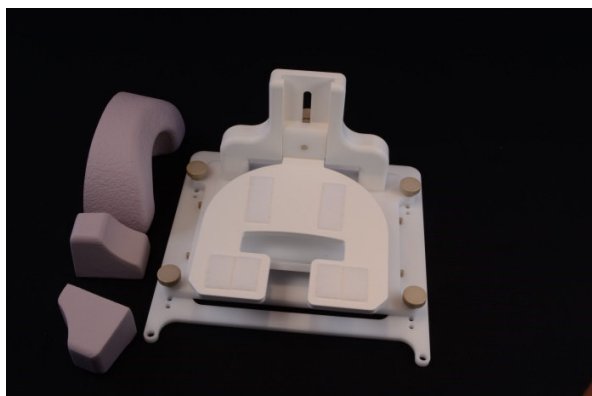
The pads of the head rest framework are mounted by hook-and-loop strips. Peel the individual pads off the frame and stick on the new ones. See the following photos for details:



Step 1: Peel of the forehead pad



Step 2: Peel of the cheek pads



Step 3: All pads removed.



Step 4: Attach the new pads to the head rest.

4.3 Returning for Service

In the event that a coil has to be returned for service, make sure that all parts of the FRU “16Ch Diagnostic Breast Coil positioning parts set” are disassembled according to this Service Manual. Otherwise the device will not fit into the dedicated packaging and might be damaged during transportation. For disassembling, follow the steps described in chapter 3 Coil Installation in reverse order.

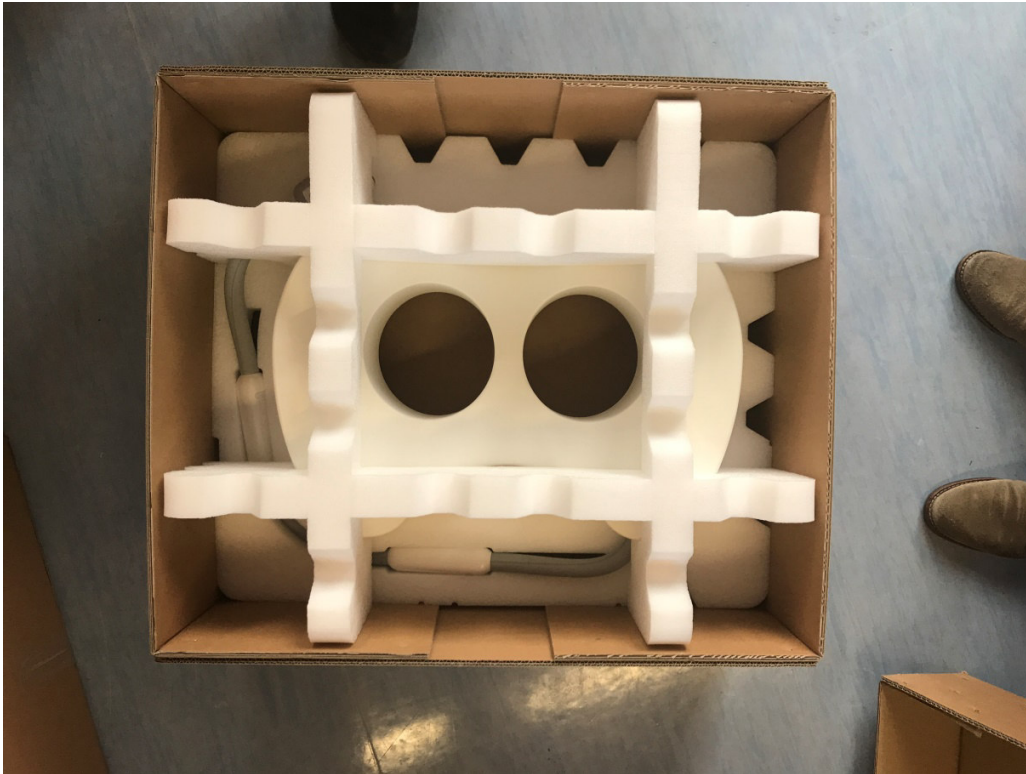


Figure 11: Dedicated packaging for 16Ch Diagnostic Breast Coil with cable

5 Troubleshooting

5.1 General notes

When encountering a problem with the 16Ch Diagnostic Breast Coils, try the following troubleshooting pointers before conducting further investigation.

- There are no bent pins on the connector.
- The coil is connected and locked to the system port. The indicator on the coil connector should be showing the green, locked symbol.
- The system detects the connected coil correctly, and shows coil without error on the screen above the bore.

5.2 Failed MCQA

When the MCQA analysis has reported a failure:

- Check the procedure (phantom, positioning, landmark...).
- Redo the MCQA.
- If the MCQA still fails, return the coil (1.5T FRU 5772249-2 or 3.0T FRU 5772248-2) for repair.